

TD 06 (Matrix)

Matrix notation used :

A matrix **M** of size **n × m**

Rows = 0..n-1

Columns = 0..m-1

Exercise 01: Sum of All Elements in a Matrix

Compute the sum of all elements in matrix M.

Exercise 02: Maximum Element in a Matrix

Find the largest value in matrix M.

Exercise 03: Count Negative Elements

Count how many numbers in the matrix are negative.

Exercise 04: Display the Diagonal (Square Matrix)

Print the main diagonal of a square matrix.

Exercise 05: Sum Each Row Separately

Compute the sum of each row and print it.

Exercise 06: Sum Each Column Separately

Compute the sum of each column.

Exercise 07: Transpose a Matrix

Create the transpose of matrix M (convert rows to columns).

Exercise 08: Check if Matrix is Square

Check if the matrix is square ($n = m$).

Exercise 09: Count Zero Elements

Count the number of zeros in matrix M.

Exercise 10: Matrix Addition (A + B)

Given two matrices A and B of the same size, compute $C = A + B$.

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Exercise 11: Matrix Product (Multiplication)

Given two matrices:

- A of size $n \times p$
- B of size $p \times m$

The product $C = A \times B$ will be size $n \times m$.

Condition for matrix multiplication:

The number of columns of A must equal the number of rows of B:

$$A(n \times p) \times B(p \times m) = C(n \times m)$$